

KOCHI, Japan

WMO: 478930

Lat: 33.5683N

Lon: 133.5490E

Elev: 16

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	32.4	13.8	11.1	42.8	16.6	12.8	43.2	14.0	47.6	12.3	47.8	3.8	280	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	91.5	77.9	90.0	77.6	88.4	77.1	80.3	86.9	79.5	86.0	78.9	85.1	6.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.5	83.7	77.8	144.5	83.2	77.1	141.0	82.7	44.0	87.2	43.1	86.2	42.4	85.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	9.9	8.6	DB	26.9	95.9	2.1	1.7	25.4	97.1	24.2	98.2	23.0	99.1	21.5	100.4
			WB	24.2	81.8	2.0	1.4	22.8	82.7	21.6	83.5	20.5	84.3	19.0	85.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	44.4	46.5	52.1	60.4	68.2	73.9	81.1	82.7	77.7	68.6	58.2	48.5
	DBStd	13.94	4.50	5.65	5.36	5.02	3.44	3.61	3.22	2.52	3.90	4.83	5.81	5.37
	HDD50	448	182	127	39	2	0	0	0	0	0	0	5	93
	HDD65	2461	638	518	400	150	10	0	0	0	0	20	214	511
	CDD50	5422	9	30	104	314	565	716	964	1015	831	577	250	47
	CDD65	1960	0	0	0	12	111	266	499	550	381	132	9	0
	CDH74	17076	0	0	1	35	416	1478	5160	6253	3112	605	16	0
	CDH80	5500	0	0	0	1	28	220	1797	2487	903	64	0	0
(14) Wind	WSAvg	4.0	4.0	4.1	4.2	4.2	4.1	3.7	3.9	4.1	4.0	3.8	3.8	3.9
(15) Precipitation	PrecAvg	104.4	2.4	3.8	6.8	10.5	10.2	14.8	12.0	12.6	14.0	7.1	5.3	2.5
	PrecMax	172.6	7.7	13.1	11.9	21.0	20.3	33.9	34.4	61.5	53.3	22.8	16.3	12.4
	PrecMin	68.5	0.0	0.8	0.6	3.4	1.9	4.7	1.6	0.5	2.2	0.9	0.3	0.0
	PrecStd	22.6	1.7	2.8	2.9	4.1	3.9	6.2	7.2	10.2	9.6	4.6	3.9	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	67.2	70.8	77.7	83.1	87.8	93.5	94.3	91.1	84.3	76.0	67.6
		MCWB	51.1	58.0	57.6	62.4	64.7	76.1	77.7	77.5	76.0	71.1	63.8	57.3
	2%	DB	58.4	62.9	67.4	74.4	80.1	84.5	91.1	92.1	88.6	81.7	73.1	63.8
		MCWB	48.5	54.8	56.0	60.8	65.1	73.5	77.9	78.1	75.4	69.2	62.1	53.2
	5%	DB	55.9	59.6	64.9	72.0	78.1	82.4	89.5	90.6	86.8	79.7	70.6	60.9
		MCWB	46.5	51.0	54.8	60.2	65.0	72.5	77.7	78.0	74.6	67.8	60.5	51.5
	10%	DB	53.5	56.7	62.3	69.9	76.2	80.6	87.7	89.1	85.0	77.5	68.1	58.1
		MCWB	45.0	47.9	53.2	59.4	64.6	71.4	77.2	77.6	74.1	66.3	59.3	49.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	61.5	62.8	68.9	73.2	78.6	81.0	81.5	80.0	76.9	69.1	61.5
		MCDB	58.5	64.7	65.9	72.5	76.2	84.0	88.3	88.6	85.8	80.7	71.5	64.4
	2%	WB	51.8	57.3	59.9	66.2	71.5	77.3	79.9	80.7	78.8	74.0	65.8	56.4
		MCDB	54.7	61.1	63.5	70.2	74.6	81.8	86.6	87.4	84.2	78.4	69.3	60.6
	5%	WB	48.7	53.2	57.7	64.2	69.8	76.0	79.2	80.0	77.9	71.5	63.3	53.7
		MCDB	53.3	57.4	62.3	68.5	73.4	80.0	85.5	86.6	83.3	75.9	67.6	57.7
	10%	WB	46.1	49.9	55.3	62.3	67.8	74.5	78.5	79.2	76.8	69.4	61.2	51.0
		MCDB	52.2	54.8	60.4	67.3	72.8	78.3	84.6	85.8	82.3	74.2	66.0	56.7
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	16.5	16.6	16.4	15.1	11.9	11.2	12.3	13.2	15.4	16.2	16.8
		MCDBR	19.0	18.2	18.9	18.6	18.2	14.2	14.1	14.4	14.8	16.8	17.1	18.9
		MCWBR	11.5	11.5	12.0	10.2	9.0	6.2	5.4	5.5	6.1	7.7	9.3	11.2
	5% WB	MCDBR	15.7	14.9	15.3	13.4	12.4	10.0	11.6	11.9	11.6	11.9	12.9	15.0
		MCWBR	11.8	11.6	12.3	9.6	8.7	5.2	5.0	5.1	5.6	7.3	9.0	11.1
(40) Clear-Sky Solar Irradiance	taub		0.424	0.479	0.547	0.578	0.582	0.596	0.591	0.560	0.520	0.480	0.443	0.414
	taud		2.077	1.959	1.811	1.792	1.845	1.885	1.940	2.011	2.087	2.109	2.116	2.121
	Ebn at Noon		239	239	233	233	234	230	230	235	238	236	232	234
	Edh at Noon		41	52	65	69	66	63	60	55	49	44	40	38
(44) All-Sky Solar Radiation	RadAvg		842	1019	1313	1531	1619	1367	1563	1600	1281	1082	843	733
	RadStd		66	75	100	167	179	170	193	166	123	114	76	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.67	+0.92	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page